

YAP, Micronesia, Federated States of

WMO: 914130

Lat: 9.497N

Lon: 138.082E

Elev: 28

StdP: 100.99

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 40308

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.7	24.0	20.8	15.5	27.0	21.4	16.1	27.0	8.4	28.3	8.0	28.4	2.1	50	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.8	33.2	27.3	32.5	27.0	32.0	26.8	28.1	31.7	27.7	31.3	27.4	31.0	4.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	30.3	26.7	22.3	30.0	26.4	21.9	29.8	90.2	31.8	88.4	31.5	87.1	31.2	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	7.2	6.5	22.5	34.1	0.7	1.8	22.0	35.4	21.6	36.4	21.2	37.4	20.7	38.7		
(5)			20.9	29.4	1.0	1.1	20.2	30.2	19.6	30.8	19.1	31.4	18.4	32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	27.7	27.7	28.0	28.3	28.5	28.3	28.0	27.9	27.9	28.1	28.2	28.0		
	DBStd	0.92	0.82	0.80	0.83	0.84	0.94	1.02	0.96	0.97	0.97	0.89	0.75	0.82		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6590	550	496	557	550	573	548	558	554	538	562	546	559		
	CDD18.3	3548	292	262	298	300	314	298	299	296	288	303	296	301		
	CDH23.3	38550	3014	2692	3161	3343	3564	3319	3244	3210	3160	3321	3271	3250		
	CDH26.7	12266	823	734	966	1126	1268	1137	1046	1012	1017	1106	1063	967		
(14) Wind	WSAvg	3.6	4.1	4.3	4.2	3.9	3.2	3.1	3.2	3.3	3.3	3.0	3.4	3.9		
(15) Precipitation	PrecAvg	3105	190	152	151	151	234	311	377	384	354	315	229	238		
	PrecMax	4032	552	488	361	405	562	813	882	716	640	562	525	659		
	PrecMin	2257	32	7	17	4	38	66	150	130	71	66	60	56		
	PrecStd	468	113	103	87	93	127	124	127	132	112	124	90	110		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	32.0	33.0	33.3	33.8	34.1	33.5	33.2	33.1	33.1	32.3	32.2		
		MCWB	26.4	26.3	26.0	26.9	27.7	27.9	27.5	27.3	27.2	27.3	27.1	27.0		
	2%	DB	31.1	31.1	31.6	32.1	32.7	32.7	32.1	32.0	32.0	32.0	31.6	31.3		
		MCWB	26.1	25.9	26.1	26.6	27.3	27.4	27.1	26.9	26.9	27.0	26.9	26.6		
	5%	DB	30.6	30.6	31.0	31.5	31.9	31.7	31.4	31.2	31.3	31.4	31.1	30.8		
		MCWB	25.9	25.7	26.0	26.4	27.0	27.0	26.8	26.6	26.7	26.8	26.7	26.4		
	10%	DB	30.0	30.1	30.4	30.8	31.1	31.0	30.7	30.5	30.7	30.8	30.7	30.3		
		MCWB	25.7	25.6	25.7	26.2	26.7	26.8	26.5	26.4	26.5	26.6	26.6	26.2		
	0.4%	WB	27.5	27.6	27.9	27.7	28.4	28.6	28.2	27.8	27.9	28.1	28.0	27.8		
		MCDB	30.8	30.5	30.5	31.4	32.2	32.3	31.7	31.8	31.5	31.6	31.1	30.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.9	26.7	27.1	27.2	27.7	27.8	27.5	27.2	27.3	27.5	27.5	27.2		
		MCDB	30.1	29.9	30.5	30.9	31.6	31.7	31.2	31.0	30.9	31.1	30.7	30.3		
	5%	WB	26.5	26.3	26.6	26.9	27.3	27.4	27.1	26.9	27.0	27.1	27.1	26.9		
		MCDB	29.6	29.5	30.0	30.5	31.1	31.0	30.7	30.5	30.6	30.6	30.4	30.0		
	10%	WB	26.1	26.0	26.1	26.5	27.0	27.0	26.7	26.6	26.6	26.7	26.8	26.6		
		MCDB	29.1	29.0	29.5	30.0	30.6	30.4	30.0	29.9	30.0	30.0	29.9	29.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.1	5.6	5.6	5.8	5.8	5.6	5.4	5.5	5.7	5.4	5.0		
		MCDBR	5.7	5.7	6.2	6.2	6.7	6.8	6.6	6.4	6.4	6.5	5.9	5.5		
		MCWBR	2.8	2.7	2.8	2.8	3.1	3.3	3.2	3.1	3.1	3.1	2.9	2.7		
	5% WB	MCDBR	5.4	5.4	5.7	5.8	6.2	6.4	6.2	5.9	5.9	6.0	5.7	5.2		
		MCWBR	2.9	2.8	3.0	2.9	3.1	3.3	3.2	3.0	3.0	3.1	3.0	2.8		
(40) Clear-Sky Solar Irradiance	taub		0.442	0.436	0.436	0.428	0.406	0.398	0.401	0.404	0.396	0.410	0.428	0.447		
	taud		2.403	2.421	2.440	2.487	2.587	2.605	2.595	2.570	2.599	2.542	2.484	2.394		
	Ebn at Noon		869	886	887	883	886	883	883	892	910	899	876	855		
	Edh at Noon		119	120	119	113	100	96	98	102	100	105	109	118		
	RadAvg		5.03	5.51	6.02	6.25	5.86	5.46	5.05	4.98	5.23	5.26	5.02	4.80		
(45) All-Sky Solar Radiation	RadStd		0.35	0.40	0.34	0.32	0.54	0.24	0.46	0.40	0.42	0.51	0.36	0.40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.30	+0.38	N/A	+0.65	+0.34	+0.38	N/A	N/A	+132	+132	
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page